



VPC TRAFFIC FLOW AND SECURITY

By Ahmed Umar Rehman | September 2026

The screenshot displays the AWS Management Console interface for a NextWork Security Group. At the top, a green notification banner states: "Security group (sg-0b41415aa21da55e5 | NextWork Security Group) was created successfully". Below this, the page title is "sg-0b41415aa21da55e5 - NextWork Security Group".

The "Details" section provides the following information:

- Security group name:** NextWork Security Group
- Security group ID:** sg-0b41415aa21da55e5
- Description:** A Security Group for the NextWork VP C
- VPC ID:** vpc-0fae0b5c00e534de9
- Owner:** 689744602638
- Inbound rules count:** 1 Permission entry
- Outbound rules count:** 1 Permission entry

Below the details, there are tabs for "Inbound rules", "Outbound rules", "Sharing", "VPC associations", "Related resources - new", and "Tags". The "Inbound rules" tab is selected, showing a table with one rule:

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sg-06c60307ad43db665	IPv4	HTTP	TCP	80

The interface also includes a left-hand navigation menu with options like "VPC", "Security Groups", "NAT gateways", "PrivateLink and Lattice", and "DNS firewall". The top right corner shows the account ID "6897-4460-2638" and the user "ahmedumar-lam-admin".

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a logically isolated virtual network in AWS, and it is useful because it allows you to control and secure how your AWS resources communicate with each other and the internet.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create and configure a virtual network with subnets, route tables, an Internet Gateway, Security Groups, and Network ACLs to control and secure network traffic.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was running into IAM permission issues, especially when my IAM user was unable to perform some VPC actions even after permissions were assigned.

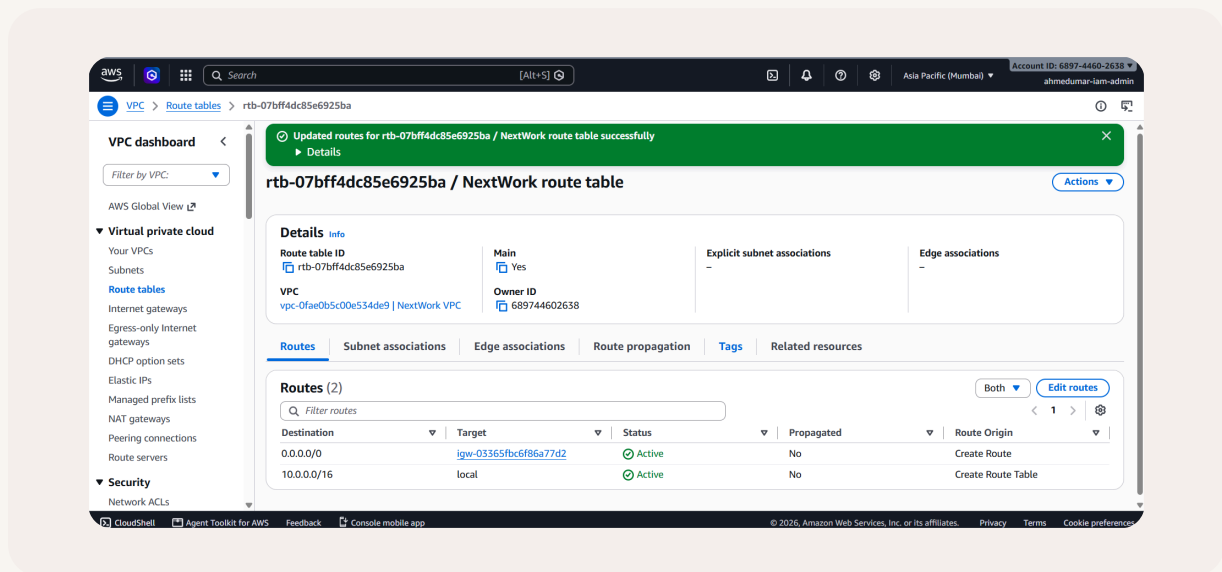
This project took me...

This project took me around 1 hour and 20 minutes

Route tables

Route tables are rules that determine where network traffic from a subnet should go.

Route tables are needed to make a subnet public because they provide a route to an Internet Gateway, allowing the subnet's resources to communicate with the internet.



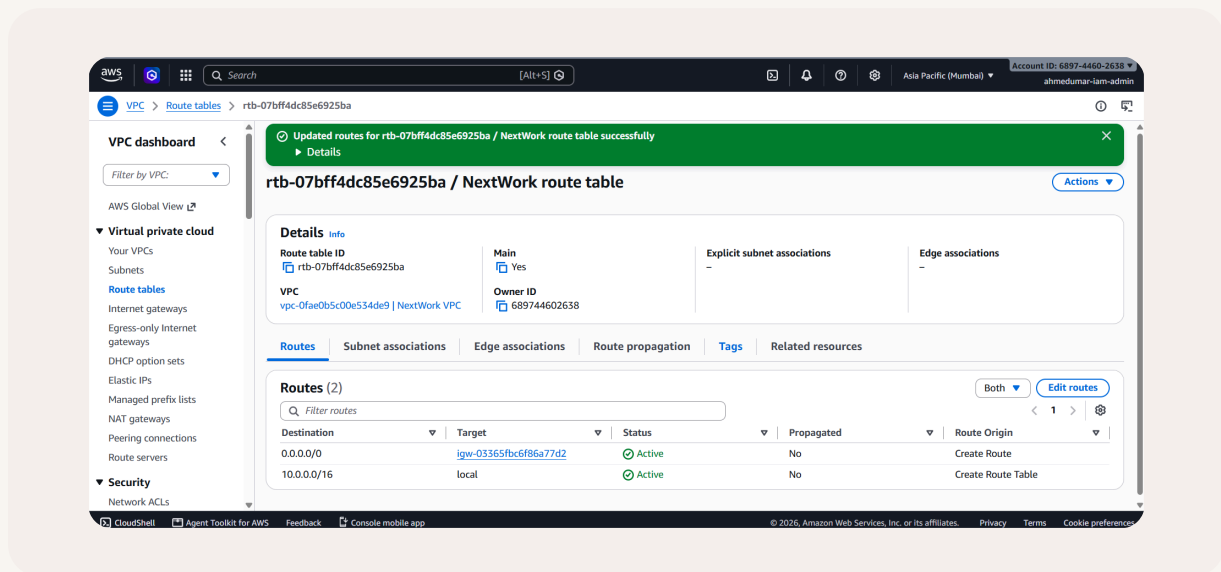
Route destination and target

Routes are defined by their destination and target, which mean

Destination = where the traffic is going.

Target = where the traffic should be sent.

The route had a destination of 0.0.0.0/0 and a target of my Internet Gateway.



Security groups

Security groups are virtual firewalls that control inbound and outbound traffic for AWS resources such as EC2 instances.

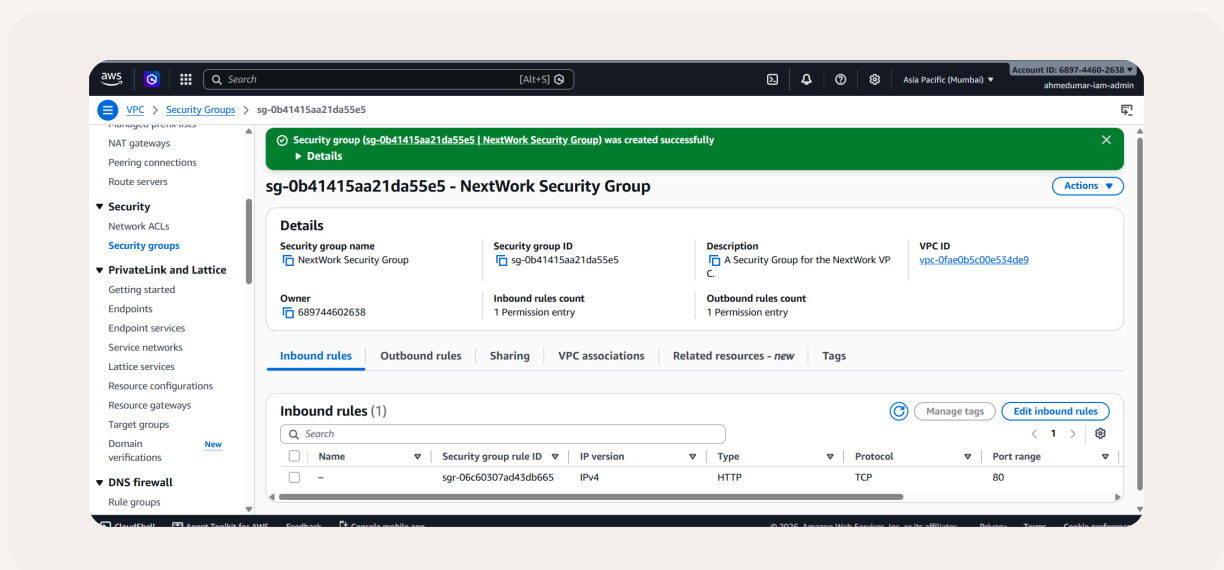
Inbound vs Outbound rules

Inbound rules are rules that control what incoming traffic is allowed to reach a resource.

I configured an inbound rule that allows specific incoming traffic to reach my resource.

Outbound rules are rules that control what outgoing traffic is allowed from a resource.

By default, my security group's outbound rule allows all outbound traffic.



Network ACLs

Network ACLs (NACLs) are subnet-level firewalls that control inbound and outbound traffic using allow and deny rules.

Security groups vs. network ACLs

The difference between a security group and a network ACL is that a security group operates at the resource level and is stateful, while a network ACL operates at the subnet level and is stateless.

Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will allow all traffic.

In contrast, a custom NACL's inbound and outbound rules are automatically set to deny all traffic.

